

Scalability & Future Plan

Date	29 June 2024
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static CSV dataset instead of a live database	New applicant data cannot be stored permanently.	High
2	Supports only single loan prediction at a time.	Not suitable for bulk loan application processing.	Medium
3	No user authentication or role-based access.	Limits security for real-world banking use.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for limited users.	Deploy on scalable cloud infrastructure with load balancing
2	Data Storage	Uses a CSV dataset.	Integrate MySQL or PostgreSQL for secure data storage.
3	Performance	Processes one prediction at a time	Optimize model inference and support batch predictions.
4	Security	No authentication implemented.	Add secure login, user authentication, and HTTPS support.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
.Phase 1	Database integration for storing applicant records	Short Term	Improves data management and persistence.
Phase 2	User authentication and role-based access control.	Medium Term	Enhances application security.
Phase 3	Prediction history and reporting dashboard.	Medium Term	Enables tracking and analysis of loan applications.
Phase 4	Explainable AI and advanced machine learning models.	Long Term	Improves prediction transparency and overall accuracy.